Relativistic Electrodynamics

Jake W. Liu

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Outline

Chapter Overview

This chapter bridges classical electromagnetism and the relativistic framework, demonstrating that Maxwell equations are inherently consistent with Einstein's special relativity. We begin by reviewing the foundational postulates and deriving the Lorentz transformation for spacetime coordinates. The corresponding transformation laws for the electric and magnetic fields are then developed, illustrating how these quantities are perceived by observers in relative motion. These ideas underlie the velocity-dependent part of Global Positioning System (GPS) timing corrections (whose gravitational part requires general relativity), particle-accelerator beam dynamics, and relativistic charged-particle radiation, highlighting the practical importance of relativity in electromagnetic applications.

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Special Relativity and Lorentz Transformation

The Lorentz transformation describes how space and time coordinates transform between inertial frames moving at constant velocity relative to each other. Its form is determined by two fundamental postulates that underlie the theory of special relativity:

(P1) The laws of physics are the same in all inertial frames (principle of relativity).

(P2) The speed of light in vacuum, c, is the same for all inertial observers, independent of the motion of the source or the observer.

Postulate (P1) by itself does not mathematically imply linearity or spacetime homogeneity. We therefore make explicit the usual background assumptions used in deriving an inertial-coordinate transformation: space and time are homogeneous, space is isotropic, and inertial worldlines are mapped to inertial worldlines. Homogeneity makes the transformation affine; because the origins coincide at $t = t' = 0$, its constant part vanishes and the transformation is linear. Isotropy implies that reversing the boost direction cannot change a scale factor that depends only on the speed $|v|$.
Consider first the standard configuration in which S' moves with velocity $v\hat{x}$ relative to S, the coordinate axes are parallel, and $|v| < c$. Rotational and reflection symmetry about the x-axis forbid the longitudinal coordinate and time from mixing with y or z.

Special Relativity and Lorentz Transformation (cont.)

The worldline of the S' origin is $x = vt$ and must map to $x' = 0$. The most general linear longitudinal relation with this property is therefore

$$x' = A(v)(x - vt) = A(v)(x - \beta ct), \qquad \beta \equiv \frac{v}{c}. \tag{5.1.1}$$

Write the still-unknown linear time transformation in dimensionally uniform form as

$$ct' = D(v)ct + F(v)x. \tag{5.1.2}$$

A light ray emitted at the common origin and traveling in the $+x$ direction obeys $x = ct$ in S and, by (P2), $x' = ct'$ in S'. Substitution into (5.1.1) gives

$$ct' = x' = A(v)c(1 - \beta)t, \qquad \text{so} \qquad D + F = A(1 - \beta).$$

For the ray traveling in the $-x$ direction, $x = -ct$ and $x' = -ct'$, so

$$ct' = -x' = A(v)c(1 + \beta)t, \qquad \text{so} \qquad D - F = A(1 + \beta).$$

Adding and subtracting these two coefficient equations yields every intermediate coefficient,

$$2D = 2A, \qquad 2F = -2A\beta, \qquad D = A, \qquad F = -A\beta.$$

Special Relativity and Lorentz Transformation (cont.)

Thus

$$ct' = A(v)(ct - \beta x). \tag{5.1.3}$$

The principle of relativity requires the inverse transformation to have the same form with v replaced by $-v$. Isotropy gives $A(-v) = A(v)$, so

$$x = A(v)(x' + vt') = A(v)(x' + \beta ct').$$

Substituting (5.1.1) and (5.1.3) into this inverse relation gives

$$\begin{aligned} x &= A^2\left[(x - \beta ct) + \beta(ct - \beta x)\right] \\ &= A^2(1 - \beta^2)x. \end{aligned}$$

Since this must hold for every x and the transformation must approach the identity continuously as $v \to 0$, the positive solution is

$$A(v) = \frac{1}{\sqrt{1 - \beta^2}} \equiv \gamma. \tag{5.1.4}$$

Special Relativity and Lorentz Transformation (cont.)

The one-dimensional Lorentz boost is therefore

$$x' = \gamma\left(x - \beta ct\right), \tag{5.1.5a}$$
$$ct' = \gamma\left(ct - \beta x\right). \tag{5.1.5b}$$

Interval invariance is now a consequence rather than an assumption. Expanding the transformed longitudinal interval gives

$$\begin{aligned}
x'^2 - c^2 t'^2 &= \gamma^2\left[(x-\beta ct)^2 - (ct-\beta x)^2\right] \\
&= \gamma^2\left[x^2 - 2\beta xct + \beta^2 c^2 t^2 - c^2 t^2 + 2\beta xct - \beta^2 x^2\right] \\
&= \gamma^2(1-\beta^2)(x^2 - c^2 t^2) \\
&= x^2 - c^2 t^2.
\end{aligned}$$

For completeness, rotational symmetry about the boost axis permits only a common transverse scale, $y' = C(v)y$ and $z' = C(v)z$. The inverse boost gives $C(v)C(-v) = 1$, while spatial isotropy gives $C(-v) = C(v)$. Hence $C^2 = 1$, and continuity at $v = 0$ selects $C = 1$: $y' = y$ and $z' = z$. The spacetime interval

$$s^2 = \vec{r}\cdot\vec{r} - c^2 t^2 \tag{5.1.6}$$

satisfies

$$s'^2 = x'^2 + y'^2 + z'^2 - c^2t'^2 = x^2 + y^2 + z^2 - c^2t^2 = s^2.$$

The same boost can be viewed as a hyperbolic rotation. Define the rapidity ϕ by $\tanh\phi = \beta$. Then

$$\cosh\phi = \frac{1}{\sqrt{1-\tanh^2\phi}} = \gamma,$$

$$\sinh\phi = \tanh\phi\cosh\phi = \gamma\beta.$$

and (5.1.5) becomes

$$x' = x\cosh\phi - ct\sinh\phi, \tag{5.1.7}$$

$$ct' = -x\sinh\phi + ct\cosh\phi. \tag{5.1.8}$$

The identity $\cosh^2\phi - \sinh^2\phi = 1$ is exactly the hyperbolic counterpart of the cancellation shown explicitly above.

Before generalizing the boost direction, it is useful to derive relativity of simultaneity and longitudinal length contraction from the transformation just obtained. Let a ruler be at rest in S' and parallel to the x'-axis. To measure its moving length in S, an observer must record the two endpoint positions simultaneously, so the selected endpoint events

Special Relativity and Lorentz Transformation (cont.)

obey $\Delta t = 0$. Although those events need not be simultaneous in S', their spatial separation $\Delta x'$ equals the ruler's proper length because both endpoints have fixed x' coordinates.

Returning to the case of a boost along the x-axis for simplicity, consider two events whose space–time coordinates in the frame S are (t_1, x_1) and (t_2, x_2), with spatial and temporal separations

$$\Delta x = x_2 - x_1, \qquad \Delta t = t_2 - t_1.$$

Using (5.1.5), the corresponding spatial and temporal separations in frame S' are

$$\Delta x' = \gamma \left(\Delta x - \beta c \Delta t\right), \tag{5.1.9a}$$

$$c\Delta t' = \gamma \left(c\Delta t - \beta \Delta x\right). \tag{5.1.9b}$$

Alternatively, since S moves with a constant velocity $-v$ relative to S', we have

$$\Delta x = \gamma \left(\Delta x' + \beta c \Delta t'\right), \tag{5.1.10a}$$

$$c\Delta t = \gamma \left(c\Delta t' + \beta \Delta x'\right). \tag{5.1.10b}$$

Special Relativity and Lorentz Transformation (cont.)

For the simultaneous endpoint events selected in S,

$$\Delta x \neq 0, \qquad \Delta t = 0.$$

Substituting these conditions into the second equation of (5.1.9) yields

$$c\Delta t' = -\gamma\beta\,\Delta x \neq 0 \qquad (\Delta t = 0). \tag{5.1.11}$$

Therefore, spatially separated events that are simultaneous in frame S are *not* simultaneous in frame S'. By the same reasoning, events simultaneous in S' at different locations are not simultaneous in S. It is useful to recast this relativity of simultaneity as a direct relation between the separations measured in S'. Still imposing $\Delta t = 0$, the first equation of (5.1.9) gives

$$\Delta x' = \gamma\,\Delta x \qquad (\Delta t = 0),$$

so the ruler's proper length $\Delta x'$ exceeds the simultaneously measured length Δx in S. Using this to eliminate Δx from (5.1.11), the temporal separation in S' becomes

$$c\Delta t' = -\gamma\beta\,\Delta x = -\beta\,\Delta x' \qquad (\Delta t = 0).$$

The two endpoint measurements, simultaneous in S, are therefore staggered in S' by an amount fixed entirely by $\Delta x'$.
The nonzero $\Delta t'$ is precisely why reciprocal longitudinal contractions do not contradict one another: the two frames use different pairs of endpoint events. From (5.1.10), the observer in S finds

$$\Delta x = \gamma \left(\Delta x' + \beta c \Delta t'\right) = \gamma \, \Delta x' \, (1 - \beta^2) = \Delta x' \sqrt{1 - \beta^2}, \tag{5.1.12}$$

which is shorter than its proper length $\Delta x'$. Conversely, an observer in S' applying the same simultaneity condition $\Delta t' = 0$ will conclude that a ruler moving with respect to S' is also contracted along the direction of motion. Therefore, each inertial frame observes the other's parallel ruler as shorter, and this mutual contraction is fully consistent with the relativity of simultaneity. Only the component of length parallel to the relative motion undergoes Lorentz contraction, while perpendicular components remain unchanged.
We now generalize to an arbitrary boost $\vec{\beta} = \vec{v}/c$. Decompose every position into parts parallel and perpendicular to $\vec{\beta}$. Since $\vec{r}\,'_\perp = \vec{r}_\perp$, spacetime mixing occurs only in the plane spanned by the time axis and the boost direction; the parallel coordinate and time therefore obey the one-dimensional boost already derived. Explicitly:

Special Relativity and Lorentz Transformation (cont.)

1. The component of $\vec{r}$ perpendicular to the velocity, $\vec{r}_\perp$, is unchanged: $\vec{r}\,'_\perp = \vec{r}_\perp$.
2. The time coordinate and the parallel spatial component transform according to the one-dimensional rules. The time transformation generalizes by replacing βx with the scalar projection $\vec{\beta} \cdot \vec{r}$, where $\vec{\beta} = \beta\hat{\beta}$.

Applying these rules, the time coordinate transformation becomes:

$$ct' = \gamma \left(ct - \vec{\beta} \cdot \vec{r} \right). \tag{5.1.13}$$

For the spatial coordinates, we have $\vec{r}\,' = \vec{r}\,'_\parallel + \vec{r}\,'_\perp$. With $\vec{r}\,'_\perp = \vec{r}_\perp$ and $\vec{r}\,'_\parallel = \gamma(\vec{r}_\parallel - \vec{\beta}ct)$, we can write:

$$\begin{aligned} \vec{r}\,' &= \vec{r}_\perp + \gamma\,(\vec{r}_\parallel - \vec{\beta}ct) \\ &= (\vec{r} - \vec{r}_\parallel) + \gamma\vec{r}_\parallel - \gamma\vec{\beta}ct \\ &= \vec{r} + (\gamma - 1)\vec{r}_\parallel - \gamma\vec{\beta}ct. \end{aligned} \tag{5.1.14}$$

Special Relativity and Lorentz Transformation (cont.)

For $\beta \neq 0$, use $\vec{r}_{\parallel} = (\vec{\beta} \cdot \vec{r})\vec{\beta}/\beta^2$ to obtain the full spatial transformation; at $\beta = 0$ its continuous limit is simply $\vec{r}' = \vec{r}$:

$$\vec{r}\,' = \vec{r} + (\gamma - 1)\frac{\vec{\beta} \cdot \vec{r}}{\beta^2}\vec{\beta} - \gamma\vec{\beta}ct. \tag{5.1.15}$$

This expression is often written more compactly using dyadic notation. If we define the symmetric tensor

$$\bar{\bar{\alpha}} = \bar{\bar{I}} + (\gamma - 1)\frac{\vec{\beta}\vec{\beta}}{\beta^2}, \tag{5.1.16}$$

where $\bar{\bar{I}}$ is the identity tensor and $\vec{\beta}\vec{\beta}$ is the outer product, the spatial transformation is:

$$\vec{r}\,' = \bar{\bar{\alpha}} \cdot \vec{r} - \gamma\vec{\beta}ct. \tag{5.1.17}$$

(5.1.13) and (5.1.17) form the complete, general Lorentz transformation for a boost in any direction $\vec{v}$.

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Lorentz Transformation of Maxwell Equations

The quantities in this section are real, instantaneous spacetime fields, not complex phasors; consequently, a square such as $|\vec{E}|^2$ means the Euclidean dot product $\vec{E}\cdot\vec{E}$ with no complex conjugation. Transformation formulas for electromagnetic field vectors follow from the Lorentz transformation for space and time. The derivation below gives

$$\begin{bmatrix} \vec{E}' \\ c\vec{B}' \end{bmatrix} = \gamma \begin{bmatrix} \bar{\bar{\alpha}}^{-1} & \bar{\bar{\beta}} \\ -\bar{\bar{\beta}} & \bar{\bar{\alpha}}^{-1} \end{bmatrix} \cdot \begin{bmatrix} \vec{E} \\ c\vec{B} \end{bmatrix} \tag{5.2.1}$$

and

$$\begin{bmatrix} c\vec{D}' \\ \vec{H}' \end{bmatrix} = \gamma \begin{bmatrix} \bar{\bar{\alpha}}^{-1} & \bar{\bar{\beta}} \\ -\bar{\bar{\beta}} & \bar{\bar{\alpha}}^{-1} \end{bmatrix} \cdot \begin{bmatrix} c\vec{D} \\ \vec{H} \end{bmatrix}. \tag{5.2.2}$$

Here, $\bar{\bar{\alpha}}^{-1}$ represents the inverse of $\bar{\bar{\alpha}}$, which is given by:

$$\bar{\bar{\alpha}}^{-1} = \bar{\bar{I}} + \left(\frac{1}{\gamma} - 1\right)\frac{\vec{\beta}\vec{\beta}}{\beta^2} = \bar{\bar{\alpha}} - \gamma\vec{\beta}\vec{\beta}. \tag{5.2.3}$$

Lorentz Transformation of Maxwell Equations (cont.)

At $\beta = 0$, expressions containing $\vec{\beta}\vec{\beta}/\beta^2$ are understood by their continuous limit: $\gamma \to 1$ and $\bar{\bar{\alpha}}, \bar{\bar{\alpha}}^{-1} \to \bar{\bar{I}}$, independent of direction. The 3×3 matrix $\bar{\bar{\beta}}$ has the explicit matrix form:

$$\bar{\bar{\beta}} = \begin{bmatrix} 0 & -\beta_z & \beta_y \\ \beta_z & 0 & -\beta_x \\ -\beta_y & \beta_x & 0 \end{bmatrix}. \tag{5.2.4}$$

For any $\vec{u} = (u_x, u_y, u_z)$, direct matrix multiplication gives

$$\bar{\bar{\beta}} \cdot \vec{u} = \begin{bmatrix} -\beta_z u_y + \beta_y u_z \\ \beta_z u_x - \beta_x u_z \\ -\beta_y u_x + \beta_x u_y \end{bmatrix} = \vec{\beta} \times \vec{u}.$$

From the operator identity

$$\bar{\bar{\beta}}^2 \cdot \vec{u} = \bar{\bar{\beta}} \cdot (\bar{\bar{\beta}} \cdot \vec{u}) = \vec{\beta} \times (\vec{\beta} \times \vec{u}) = \vec{\beta}\vec{\beta} \cdot \vec{u} - \beta^2 \vec{u},$$

it follows that:

$$\bar{\bar{\beta}}^2 = \vec{\beta}\vec{\beta} - \beta^2 \bar{\bar{I}}. \tag{5.2.5}$$

While both $\bar{\bar{\alpha}}$ and $\bar{\bar{\alpha}}^{-1}$ are symmetric, $\bar{\bar{\beta}}$ is skew-symmetric.

Both equalities asserted in (5.2.3) deserve a word, since the second one is not a purely algebraic identity. The key observation is that the dyad $\vec{\beta}\vec{\beta}/\beta^2$ is idempotent: because $\vec{\beta}\cdot\vec{\beta}=\beta^2$,

$$\left(\vec{\beta}\vec{\beta}\right)\cdot\left(\vec{\beta}\vec{\beta}\right)=\vec{\beta}\left(\vec{\beta}\cdot\vec{\beta}\right)\vec{\beta}=\beta^2\,\vec{\beta}\vec{\beta}.$$

To confirm that the first form is indeed the inverse, we multiply it by $\bar{\bar{\alpha}}$ and use this rule on the cross term:

$$\begin{aligned}\bar{\bar{\alpha}}\cdot\bar{\bar{\alpha}}^{-1}&=\left[\bar{\bar{I}}+(\gamma-1)\frac{\vec{\beta}\vec{\beta}}{\beta^2}\right]\cdot\left[\bar{\bar{I}}+\left(\tfrac{1}{\gamma}-1\right)\frac{\vec{\beta}\vec{\beta}}{\beta^2}\right]\\&=\bar{\bar{I}}+\left[(\gamma-1)+\left(\tfrac{1}{\gamma}-1\right)+(\gamma-1)\left(\tfrac{1}{\gamma}-1\right)\right]\frac{\vec{\beta}\vec{\beta}}{\beta^2}.\end{aligned}$$

The bracketed coefficient is $(\gamma-1)+(\frac{1}{\gamma}-1)+(1-\gamma-\frac{1}{\gamma}+1)=0$, so $\bar{\bar{\alpha}}\cdot\bar{\bar{\alpha}}^{-1}=\bar{\bar{I}}$, as required.

The second equality, $\bar{\bar{\alpha}}^{-1}=\bar{\bar{\alpha}}-\gamma\vec{\beta}\vec{\beta}$, is more subtle. Writing the right-hand side out,

$$\bar{\bar{\alpha}}-\gamma\vec{\beta}\vec{\beta}=\bar{\bar{I}}+\left[(\gamma-1)-\gamma\beta^2\right]\frac{\vec{\beta}\vec{\beta}}{\beta^2},$$

Lorentz Transformation of Maxwell Equations (cont.)

so matching the coefficient of $\vec{\beta}\vec{\beta}/\beta^2$ against (5.2.3) demands

$$(\gamma - 1) - \gamma\beta^2 = \frac{1}{\gamma} - 1 \quad \Longleftrightarrow \quad \gamma^2\beta^2 = \gamma^2 - 1.$$

This is precisely the Lorentz-factor relation (5.1.4), $\gamma = 1/\sqrt{1-\beta^2}$ (equivalently $\beta^2 = 1 - 1/\gamma^2$). The second form is therefore not a general dyadic identity; it holds only on account of the kinematic constraint linking γ and β, and it is this substitution that lets $\bar{\bar{\alpha}} - \gamma\vec{\beta}\vec{\beta}$ collapse to $\bar{\bar{\alpha}}^{-1}$ in the field-transformation manipulations that follow.
To prove this, we first deduce the Lorentz transformation for space-time derivatives. This derivation utilizes the chain rule in differentiation:

$$\partial_{ct} = \left[\partial_{ct}(ct')\right]\partial_{ct'} + \left[\partial_{ct}x'_i\right]\partial_{x'_i}, \tag{5.2.6}$$

$$\partial_{x_i} = \left[\partial_{x_i}(ct')\right]\partial_{ct'} + \left[\partial_{x_i}x'_j\right]\partial_{x'_j}. \tag{5.2.7}$$

Substituting the Lorentz transformation and acknowledging the symmetrical nature of $\bar{\bar{\alpha}}$, we evaluate these coefficients. In component form with the Einstein summation convention, (5.1.13) and (5.1.17) read $ct' = \gamma\,(ct - \beta_k x_k)$ and $x'_i = \alpha_{ik}x_k - \gamma\beta_i\,ct$,

Lorentz Transformation of Maxwell Equations (cont.)

where $\alpha_{ik} = \delta_{ik} + (\gamma - 1)\beta_i\beta_k/\beta^2$ are the entries of $\bar{\bar{\alpha}}$. Treating ct and the x_k as independent variables, the four Jacobian factors required in (5.2.6) and (5.2.7) are

$$\partial_{ct}(ct') = \gamma, \qquad \partial_{ct}x_i' = -\gamma\beta_i,$$

$$\partial_{x_i}(ct') = -\gamma\beta_i, \qquad \partial_{x_i}x_j' = \delta_{ij} + (\gamma - 1)\frac{\beta_i\beta_j}{\beta^2} = \alpha_{ij}.$$

Substituting these into the chain rule (5.2.6)–(5.2.7) gives

$$\partial_{ct} = \gamma\,\partial_{ct'} - \gamma\beta_i\,\partial_{x_i'}, \qquad \partial_{x_i} = -\gamma\beta_i\,\partial_{ct'} + \alpha_{ij}\,\partial_{x_j'}.$$

Because $\bar{\bar{\alpha}}$ is symmetric, $\alpha_{ij} = \alpha_{ji}$, so $\alpha_{ij}\,\partial_{x_j'} = \left(\bar{\bar{\alpha}} \cdot \vec{\nabla}'\right)_i$ and $\beta_i\,\partial_{x_i'} = \vec{\beta} \cdot \vec{\nabla}'$. Collecting the three spatial components into $\vec{\nabla}$ then yields the vector forms

$$\partial_{ct} = \gamma\partial_{ct'} - \gamma\vec{\beta} \cdot \vec{\nabla}', \tag{5.2.8}$$

$$\vec{\nabla} = \bar{\bar{\alpha}} \cdot \vec{\nabla}' - \gamma\vec{\beta}\partial_{ct'}. \tag{5.2.9}$$

Lorentz Transformation of Maxwell Equations (cont.)

To derive the transformation laws for all field vectors, we substitute (5.2.8) and (5.2.9) into the Maxwell equations in the S frame, requiring them to maintain the same form in the S' frame.
Let us first incorporate (5.2.8) and (5.2.9) into Gauss' magnetic field law and Faraday's law:

$$\left(\bar{\bar{\alpha}} \cdot \vec{\nabla}' - \gamma\vec{\beta}\partial_{ct'}\right) \cdot c\vec{B} = 0, \tag{5.2.10}$$

$$\left(\bar{\bar{\alpha}} \cdot \vec{\nabla}' - \gamma\vec{\beta}\partial_{ct'}\right) \times \vec{E} + \gamma\left(\partial_{ct'} - \vec{\beta} \cdot \vec{\nabla}'\right) c\vec{B} = 0. \tag{5.2.11}$$

To determine the transformation laws for $\vec{E}'$ and $c\vec{B}'$, our objective is to express (5.2.10) and (5.2.11) in the forms:

$$\vec{\nabla}' \cdot c\vec{B}' = 0, \tag{5.2.12}$$

$$\vec{\nabla}' \times \vec{E}' + \partial_{ct'} c\vec{B}' = 0. \tag{5.2.13}$$

Consider γ(5.2.10)$+\gamma\vec{\beta}\cdot$(5.2.11):

$$\gamma\left[\left(\bar{\bar{\alpha}} \cdot \vec{\nabla}'\right) \cdot c\vec{B} + \vec{\beta} \cdot \left(\bar{\bar{\alpha}} \cdot \vec{\nabla}'\right) \times \vec{E} - \gamma(\vec{\beta} \cdot \vec{\nabla}')\left(\vec{\beta} \cdot c\vec{B}\right)\right] = 0.$$

Lorentz Transformation of Maxwell Equations (cont.)

Using

$$\vec{\beta} \cdot [(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \times \vec{E}] = \vec{\beta} \cdot \left[\left(\vec{\nabla}' + (\gamma - 1) \frac{\vec{\beta} \cdot \vec{\nabla}'}{\beta^2} \vec{\beta} \right) \times \vec{E} \right]$$
$$= \vec{\beta} \cdot (\vec{\nabla}' \times \vec{E}) = -\vec{\nabla}' \cdot (\vec{\beta} \times \vec{E}),$$

where the term proportional to $\vec{\beta} \times \vec{E}$ is perpendicular to $\vec{\beta}$, and using (5.2.3), we can show that:

$$\vec{\nabla}' \cdot [\gamma \bar{\bar{\alpha}}^{-1} \cdot c\vec{B} - \gamma \vec{\beta} \times \vec{E}] = 0. \tag{5.2.14}$$

Similarly, consider $\bar{\bar{\alpha}} \cdot$(5.2.11)$+\gamma\vec{\beta}$(5.2.10):

$$\bar{\bar{\alpha}} \cdot \left[\left(\bar{\bar{\alpha}} \cdot \vec{\nabla}' \right) \times \vec{E} \right] - \gamma \bar{\bar{\alpha}} \cdot \left[\left(\vec{\beta} \cdot \vec{\nabla}' \right) c\vec{B} \right] + \gamma \vec{\beta} \left[\left(\bar{\bar{\alpha}} \cdot \vec{\nabla}' \right) \cdot c\vec{B} \right]$$
$$+ \partial_{ct'} \left[-\gamma \bar{\bar{\alpha}} \cdot \left(\vec{\beta} \times \vec{E} \right) + \gamma \left(\bar{\bar{\alpha}} \cdot c\vec{B} \right) - \gamma^2 \vec{\beta}\vec{\beta} \cdot c\vec{B} \right] = 0.$$

We next apply the two identities

$$\bar{\bar{\alpha}} \cdot [(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \times \vec{E}] = \vec{\nabla}' \times (\gamma \bar{\bar{\alpha}}^{-1} \cdot \vec{E})$$

Lorentz Transformation of Maxwell Equations (cont.)

and

$$\begin{aligned}
&\vec{\beta}\Big[(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \cdot c\vec{B}\Big] - \bar{\bar{\alpha}} \cdot \Big[(\vec{\beta} \cdot \vec{\nabla}')c\vec{B}\Big] \\
&\quad = \vec{\beta}(\vec{\nabla}' \cdot c\vec{B}) - (\vec{\beta} \cdot \vec{\nabla}')c\vec{B} = \vec{\nabla}' \times (\vec{\beta} \times c\vec{B}).
\end{aligned}$$

The general cofactor rule for an invertible tensor $\bar{\bar{A}}$ is

$$(\bar{\bar{A}} \cdot \vec{a}) \times (\bar{\bar{A}} \cdot \vec{b}) = (\det \bar{\bar{A}})\, \bar{\bar{A}}^{-T} \cdot (\vec{a} \times \vec{b}).$$

The inverse transpose is essential for a nonsymmetric tensor. Here $\bar{\bar{\alpha}}$ is symmetric, so $\bar{\bar{\alpha}}^{-T} = \bar{\bar{\alpha}}^{-1}$, and $\det \bar{\bar{\alpha}} = \gamma$. (The tensor $\bar{\bar{\alpha}} = \bar{\bar{I}} + (\gamma - 1)\vec{\beta}\vec{\beta}/\beta^2$ has eigenvalue γ along $\hat{\beta}$ and unity transverse to it, the former being fixed by $\gamma^2(1 - \beta^2) = 1$, so its determinant is γ.) The boost is constant, so $\bar{\bar{\alpha}}$ commutes with the spatial derivatives. Writing $\vec{u} = \bar{\bar{\alpha}}^{-1} \cdot \vec{E}$, so that $\vec{E} = \bar{\bar{\alpha}} \cdot \vec{u}$, and applying the rule with $\vec{a} = \vec{\nabla}'$ gives every tensor factor explicitly:

$$\begin{aligned}
\bar{\bar{\alpha}} \cdot \Big[(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \times \vec{E}\Big] &= \bar{\bar{\alpha}} \cdot \Big[\gamma \bar{\bar{\alpha}}^{-1} \cdot (\vec{\nabla}' \times \vec{u})\Big] \\
&= \gamma\, \vec{\nabla}' \times \vec{u} \\
&= \vec{\nabla}' \times (\gamma \bar{\bar{\alpha}}^{-1} \cdot \vec{E}).
\end{aligned}$$

Lorentz Transformation of Maxwell Equations (cont.)

For clarity, the second identity follows without any suppressed tensor contraction. Put $q = \vec{\beta} \cdot \vec{\nabla}'$. Since $\bar{\bar{\alpha}} \cdot \vec{\nabla}' = \vec{\nabla}' + (\gamma - 1)\vec{\beta} q/\beta^2$, its first term is

$$\vec{\beta}(\vec{\nabla}' \cdot c\vec{B}) + \frac{\gamma - 1}{\beta^2}\vec{\beta}\, q(\vec{\beta} \cdot c\vec{B}),$$

whereas its second term is

$$q(c\vec{B}) + \frac{\gamma - 1}{\beta^2}\vec{\beta}\, q(\vec{\beta} \cdot c\vec{B}).$$

The terms proportional to $\gamma - 1$ cancel. The remainder is $\vec{\beta}(\vec{\nabla}' \cdot c\vec{B}) - q(c\vec{B})$, which equals $\vec{\nabla}' \times (\vec{\beta} \times c\vec{B})$ for constant $\vec{\beta}$. Thus the remaining two spatial terms carry the common factor γ and combine as

$$\begin{aligned} &\gamma\vec{\beta}\big[(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \cdot c\vec{B}\big] - \gamma\, \bar{\bar{\alpha}} \cdot \big[(\vec{\beta} \cdot \vec{\nabla}')c\vec{B}\big] \\ &= \gamma\, \vec{\nabla}' \times (\vec{\beta} \times c\vec{B}) = \vec{\nabla}' \times (\gamma\vec{\beta} \times c\vec{B}). \end{aligned}$$

Adding the two displays collects the entire spatial part into the single curl $\vec{\nabla}' \times (\gamma\bar{\bar{\alpha}}^{-1} \cdot \vec{E} + \gamma\vec{\beta} \times c\vec{B})$. For the term in the $\partial_{ct'}$ bracket of the intermediate, we use

$\bar{\bar{\alpha}}^{-1} = \bar{\bar{\alpha}} - \gamma\vec{\beta}\vec{\beta}$ from (5.2.3) on the two $c\vec{B}$ terms, while $\bar{\bar{\alpha}}$ acts as the identity on $\vec{\beta} \times \vec{E}$ (a vector transverse to $\vec{\beta}$),

$$\begin{aligned}
&-\gamma\,\bar{\bar{\alpha}}\cdot(\vec{\beta}\times\vec{E}) + \gamma\,\bar{\bar{\alpha}}\cdot c\vec{B} - \gamma^2\vec{\beta}\vec{\beta}\cdot c\vec{B} \\
&= -\gamma\,\vec{\beta}\times\vec{E} + \gamma\left(\bar{\bar{\alpha}} - \gamma\vec{\beta}\vec{\beta}\right)\cdot c\vec{B} \\
&= \gamma\bar{\bar{\alpha}}^{-1}\cdot c\vec{B} - \gamma\vec{\beta}\times\vec{E}.
\end{aligned}$$

Substituting these collapsed spatial and time-derivative parts back into the intermediate yields:

$$\vec{\nabla}'\times\left(\gamma\bar{\bar{\alpha}}^{-1}\cdot\vec{E} + \gamma\vec{\beta}\times c\vec{B}\right) + \partial_{ct'}\left(\gamma\bar{\bar{\alpha}}^{-1}\cdot c\vec{B} - \gamma\vec{\beta}\times\vec{E}\right) = 0. \tag{5.2.15}$$

By comparing (5.2.12), (5.2.13), (5.2.14), and (5.2.15), we derive the transformation formulas for $\vec{E}$ and $\vec{B}$:

$$\vec{E}' = \gamma\bar{\bar{\alpha}}^{-1}\cdot\vec{E} + \gamma\vec{\beta}\times c\vec{B}, \tag{5.2.16}$$

$$c\vec{B}' = \gamma\bar{\bar{\alpha}}^{-1}\cdot c\vec{B} - \gamma\vec{\beta}\times\vec{E}. \tag{5.2.17}$$

We can now express the Lorentz transformation formulas for $\vec{E}$ and $c\vec{B}$ as in (5.2.1).

Lorentz Transformation of Maxwell Equations (cont.)

The inhomogeneous Maxwell equations also require the transformation of charge and current. We derive those source transformations next and then carry the $\vec{D}, \vec{H}$ algebra through explicitly, rather than leaving the complementary half of Maxwell's equations unstated.

Consider the charge conservation equation. Transforming from frame S to S' yields:

$$\left(\bar{\bar{\alpha}} \cdot \vec{\nabla}' - \gamma\vec{\beta}\partial_{ct'}\right) \cdot \vec{J} + \gamma\left(\partial_{ct'} - \vec{\beta} \cdot \vec{\nabla}'\right) c\rho = 0. \tag{5.2.18}$$

Consequently, the charge conservation equation is Lorentz-covariant. Equation (5.2.18) follows by substituting (5.2.8) and (5.2.9) into the continuity equation $\vec{\nabla} \cdot \vec{J} + \partial_{ct}(c\rho) = 0$ in frame S. We now rewrite it in the manifestly covariant form $\vec{\nabla}' \cdot \vec{J}' + \partial_{ct'}(c\rho') = 0$. To do so, collect the terms acted on by $\vec{\nabla}'$ and, separately, those acted on by $\partial_{ct'}$. Since $\bar{\bar{\alpha}}$ is symmetric, $(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \cdot \vec{J} = \vec{\nabla}' \cdot (\bar{\bar{\alpha}} \cdot \vec{J})$, and $(\vec{\beta} \cdot \vec{\nabla}')\, c\rho = \vec{\nabla}' \cdot (\vec{\beta}\, c\rho)$, the spatial part becomes

$$\begin{aligned}&(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \cdot \vec{J} - \gamma(\vec{\beta} \cdot \vec{\nabla}')\, c\rho \\ &\quad = \vec{\nabla}' \cdot \left(\bar{\bar{\alpha}} \cdot \vec{J} - \gamma\vec{\beta}\, c\rho\right),\end{aligned}$$

while the temporal part collects under the common operator $\partial_{ct'}$,

$$-\gamma\vec{\beta}\cdot\partial_{ct'}\vec{J}+\gamma\,\partial_{ct'}\,c\rho=\partial_{ct'}\left[\gamma\left(c\rho-\vec{\beta}\cdot\vec{J}\right)\right].$$

Adding the two pieces, (5.2.18) reads

$$\vec{\nabla}'\cdot\left(\bar{\bar{\alpha}}\cdot\vec{J}-\gamma\vec{\beta}\,c\rho\right)+\partial_{ct'}\left[\gamma\left(c\rho-\vec{\beta}\cdot\vec{J}\right)\right]=0,$$

Under the standard requirement that the sources transform locally and linearly, without derivative-dependent additions, this expression takes the primed continuity-equation form when $\vec{J}'$ and $c\rho'$ are identified with the bracketed combinations:

$$c\rho'=\gamma\,\left(c\rho-\vec{\beta}\cdot\vec{J}\right), \tag{5.2.19}$$

$$\vec{J}'=\bar{\bar{\alpha}}\cdot\vec{J}-\gamma\vec{\beta}c\rho. \tag{5.2.20}$$

We can now complete the promised derivation of (5.2.2). In S, Ampère–Maxwell and electric Gauss laws are

$$\vec{\nabla}\times\vec{H}-\partial_{ct}(c\vec{D})=\vec{J},\qquad \vec{\nabla}\cdot c\vec{D}=c\rho.$$

Lorentz Transformation of Maxwell Equations (cont.)

Substitution of (5.2.8) and (5.2.9) gives

$$(\bar{\bar{\alpha}} \cdot \vec{\nabla}' - \gamma\vec{\beta}\partial_{ct'}) \cdot c\vec{D} = c\rho, \tag{5.2.21}$$

$$(\bar{\bar{\alpha}} \cdot \vec{\nabla}' - \gamma\vec{\beta}\partial_{ct'}) \times \vec{H} - \gamma(\partial_{ct'} - \vec{\beta} \cdot \vec{\nabla}')c\vec{D} = \vec{J}. \tag{5.2.22}$$

Take γ times (5.2.21) minus $\gamma\vec{\beta}\cdot$(5.2.22). The two terms containing $\vec{\beta} \cdot \partial_{ct'} c\vec{D}$ cancel, leaving

$$\gamma\Big[(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \cdot c\vec{D} - \gamma(\vec{\beta} \cdot \vec{\nabla}')(\vec{\beta} \cdot c\vec{D}) \\ - \vec{\beta} \cdot (\vec{\nabla}' \times \vec{H})\Big] = \gamma(c\rho - \vec{\beta} \cdot \vec{J}).$$

Using $\bar{\bar{\alpha}}^{-1} = \bar{\bar{\alpha}} - \gamma\vec{\beta}\vec{\beta}$ and $\vec{\nabla}' \cdot (\vec{\beta} \times \vec{H}) = -\vec{\beta} \cdot (\vec{\nabla}' \times \vec{H})$ for constant $\vec{\beta}$, this is

$$\vec{\nabla}' \cdot \left(\gamma\bar{\bar{\alpha}}^{-1} \cdot c\vec{D} + \gamma\vec{\beta} \times \vec{H}\right) = \gamma(c\rho - \vec{\beta} \cdot \vec{J}) = c\rho'.$$

It therefore has the primed Gauss-law form when

$$c\vec{D}' = \gamma\bar{\bar{\alpha}}^{-1} \cdot c\vec{D} + \gamma\vec{\beta} \times \vec{H}.$$

Lorentz Transformation of Maxwell Equations (cont.)

For Ampère's law, take $\bar{\bar{\alpha}}\cdot$(5.2.22) minus $\gamma\vec{\beta}$(5.2.21). The already-proved cofactor identity gives

$$\bar{\bar{\alpha}} \cdot [(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \times \vec{H}] = \vec{\nabla}' \times (\gamma \bar{\bar{\alpha}}^{-1} \cdot \vec{H}),$$

while the second spatial identity proved above gives

$$\gamma\bar{\bar{\alpha}} \cdot [(\vec{\beta} \cdot \vec{\nabla}') c\vec{D}] - \gamma\vec{\beta}[(\bar{\bar{\alpha}} \cdot \vec{\nabla}') \cdot c\vec{D}] = -\vec{\nabla}' \times (\gamma\vec{\beta} \times c\vec{D}).$$

The time-derivative terms combine step by step as

$$\begin{aligned}
&-\gamma\bar{\bar{\alpha}} \cdot (\vec{\beta} \times \partial_{ct'} \vec{H}) - \gamma\bar{\bar{\alpha}} \cdot \partial_{ct'} c\vec{D} + \gamma^2 \vec{\beta}\vec{\beta} \cdot \partial_{ct'} c\vec{D} \\
&= -\partial_{ct'} \left[\gamma\vec{\beta} \times \vec{H} + \gamma(\bar{\bar{\alpha}} - \gamma\vec{\beta}\vec{\beta}) \cdot c\vec{D}\right] \\
&= -\partial_{ct'} \left[\gamma\vec{\beta} \times \vec{H} + \gamma\bar{\bar{\alpha}}^{-1} \cdot c\vec{D}\right].
\end{aligned}$$

The transformed equation is consequently

$$\begin{aligned}
\vec{\nabla}' \times \left(\gamma\bar{\bar{\alpha}}^{-1} \cdot \vec{H} - \gamma\vec{\beta} \times c\vec{D}\right) - \partial_{ct'} \left(\gamma\bar{\bar{\alpha}}^{-1} \cdot c\vec{D} + \gamma\vec{\beta} \times \vec{H}\right) \\
= \bar{\bar{\alpha}} \cdot \vec{J} - \gamma\vec{\beta} c\rho = \vec{J}'.
\end{aligned}$$

Lorentz Transformation of Maxwell Equations (cont.)

Thus

$$\vec{H}' = \gamma \bar{\bar{\alpha}}^{-1} \cdot \vec{H} - \gamma \vec{\beta} \times c\vec{D},$$

and together with the expression for $c\vec{D}'$ this is precisely (5.2.2). These transformations concern the electromagnetic excitation fields; a material law such as $\vec{D} = \epsilon \vec{E}$ in the material rest frame generally does not retain the same scalar form in a moving frame.

If a nonzero charge distribution is stationary in S ($\vec{J} = 0$), then for a nonzero boost $\vec{\beta}$ it has the nonzero transformed current $\vec{J}' = -\gamma \vec{\beta} c \rho$. Conversely, a current contributes to the transformed charge density only through its component along the boost: the term is $-\gamma \vec{\beta} \cdot \vec{J}/c$ in ρ'.

For the transformation of $\vec{E}$ and $c\vec{B}$ as described in (5.2.1), we decompose the field vectors into components parallel and perpendicular to the velocity $\vec{v}$. It is noteworthy that the field components parallel to the velocity remain invariant:

$$\vec{E}'_{||} = \vec{E}_{||}, \tag{5.2.23}$$

$$\vec{B}'_{||} = \vec{B}_{||}, \tag{5.2.24}$$

$$\vec{D}'_{||} = \vec{D}_{||}, \tag{5.2.25}$$

$$\vec{H}'_{||} = \vec{H}_{||}. \tag{5.2.26}$$

Lorentz Transformation of Maxwell Equations (cont.)

Conversely, the perpendicular components transform as follows:

$$\vec{E}'_{\perp} = \gamma\left(\vec{E}_{\perp} + \vec{\beta}\times c\vec{B}_{\perp}\right), \tag{5.2.27}$$

$$c\vec{B}'_{\perp} = \gamma\left(c\vec{B}_{\perp} - \vec{\beta}\times \vec{E}_{\perp}\right), \tag{5.2.28}$$

$$c\vec{D}'_{\perp} = \gamma\left(c\vec{D}_{\perp} + \vec{\beta}\times \vec{H}_{\perp}\right), \tag{5.2.29}$$

$$\vec{H}'_{\perp} = \gamma\left(\vec{H}_{\perp} - \vec{\beta}\times c\vec{D}_{\perp}\right). \tag{5.2.30}$$

These results follow by projecting the general transforms (5.2.16) and (5.2.17) onto the directions parallel and perpendicular to $\vec{\beta}$. The decisive observation is how $\bar{\bar{\alpha}}^{-1}$ acts on each component. Since $\frac{\vec{\beta}\vec{\beta}}{\beta^2}$ is the projector onto $\hat{\beta}$, it leaves $\vec{E}_{||}$ unchanged and annihilates $\vec{E}_{\perp}$; hence (5.2.3) gives

$$\bar{\bar{\alpha}}^{-1}\cdot\vec{E}_{||} = \vec{E}_{||} + \left(\tfrac{1}{\gamma} - 1\right)\vec{E}_{||} = \tfrac{1}{\gamma}\,\vec{E}_{||}, \qquad \bar{\bar{\alpha}}^{-1}\cdot\vec{E}_{\perp} = \vec{E}_{\perp}.$$

That is, $\bar{\bar{\alpha}}^{-1}$ scales the parallel part by $1/\gamma$ and acts as the identity on the perpendicular part. The cross-product term is purely transverse, since $\vec{\beta} \times c\vec{B}$ is orthogonal to $\vec{\beta}$ and only $c\vec{B}_\perp$ survives the product:

$$\left(\vec{\beta} \times c\vec{B}\right)_{||} = \vec{0}, \qquad \vec{\beta} \times c\vec{B} = \vec{\beta} \times c\vec{B}_\perp .$$

Taking the parallel part of (5.2.16), the cross term drops out and the γ factors cancel against the $1/\gamma$ above,

$$\vec{E}'_{||} = \gamma\, \bar{\bar{\alpha}}^{-1} \cdot \vec{E}_{||} = \gamma \cdot \tfrac{1}{\gamma}\, \vec{E}_{||} = \vec{E}_{||},$$

which is (5.2.23); the identical cancellation in (5.2.17) yields (5.2.24). Taking instead the perpendicular part, $\bar{\bar{\alpha}}^{-1}$ acts as the identity and the full cross term is retained,

$$\begin{aligned} \vec{E}'_\perp &= \gamma\, \bar{\bar{\alpha}}^{-1} \cdot \vec{E}_\perp + \gamma\, \vec{\beta} \times c\vec{B}_\perp \\ &= \gamma \left(\vec{E}_\perp + \vec{\beta} \times c\vec{B}_\perp\right), \end{aligned}$$

recovering (5.2.27), and likewise (5.2.17) gives (5.2.28). The same projection applied to (5.2.2) produces the corresponding $\vec{D}$ and $\vec{H}$ relations. This behavior contrasts with a pure boost of spatial coordinates, for which the perpendicular position components are unchanged.

According to (5.2.27), a field that is purely magnetic in S generally has an electric component in S'. For a conductor at rest in S', this transformed electric field participates in the charge redistribution associated with motional electromotive force; whether a measurable terminal voltage results also depends on the conductor geometry, circuit path, and steady-state charge distribution, so a voltage does not follow from the field transformation alone. Conversely, by (5.2.28), a field that is purely electric in one frame generally has a magnetic component in a transversely moving frame. Thus an electron's rest-frame electrostatic field is generally accompanied by a magnetic field for observers relative to whom the electron moves; at events where the electric field is parallel to the boost, the cross product and hence that magnetic component vanish.

Now, let us find relations that are invariant under the Lorentz transformation. We first designate the 6×6 matrix appearing in (5.2.2) and (5.2.1) as $\bar{\bar{\Lambda}}$:

$$\bar{\bar{\Lambda}}(\vec{\beta}) = \gamma \begin{bmatrix} \bar{\bar{\alpha}}^{-1} & \bar{\bar{\beta}} \\ -\bar{\bar{\beta}} & \bar{\bar{\alpha}}^{-1} \end{bmatrix}. \tag{5.2.31}$$

The inverse transformation reverses the boost. To verify this algebraically rather than assert it, abbreviate $P = \bar{\bar{\alpha}}^{-1}$ and $Q = \bar{\bar{\beta}}$. These tensors commute, and direct block multiplication gives

$$\begin{aligned}\bar{\bar{\Lambda}}(\vec{\beta})\bar{\bar{\Lambda}}(-\vec{\beta}) &= \gamma^2 \begin{bmatrix} P & Q \\ -Q & P \end{bmatrix} \begin{bmatrix} P & -Q \\ Q & P \end{bmatrix} \\ &= \gamma^2 \begin{bmatrix} P^2 + Q^2 & -PQ + QP \\ -QP + PQ & P^2 + Q^2 \end{bmatrix}.\end{aligned}$$

The off-diagonal blocks vanish. From (5.2.3) and (5.2.5),

$$P^2 = \bar{\bar{I}} + \left(\frac{1}{\gamma^2} - 1\right)\frac{\vec{\beta}\vec{\beta}}{\beta^2}, \qquad Q^2 = \vec{\beta}\vec{\beta} - \beta^2\bar{\bar{I}},$$

so $\gamma^2(P^2 + Q^2) = \bar{\bar{I}}$ after using $1 - \beta^2 = 1/\gamma^2$. The product is therefore the 6×6 identity, proving

$$\bar{\bar{\Lambda}}^{-1}(\vec{\beta}) = \bar{\bar{\Lambda}}(-\vec{\beta}) = \gamma \begin{bmatrix} \bar{\bar{\alpha}}^{-1} & -\bar{\bar{\beta}} \\ \bar{\bar{\beta}} & \bar{\bar{\alpha}}^{-1} \end{bmatrix}. \tag{5.2.32}$$

Lorentz Transformation of Maxwell Equations (cont.)

From a physical perspective, inverting a pure Lorentz transformation is equivalent to reversing the direction of the velocity.

We further investigate some properties of the $\bar{\bar{\Lambda}}$ matrix. Given that $\bar{\bar{\alpha}}$ is symmetric and $\bar{\bar{\beta}}$ is skew-symmetric, we find that:

$$\bar{\bar{\Lambda}}^T = \gamma \begin{bmatrix} (\bar{\bar{\alpha}}^{-1})^T & (-\bar{\bar{\beta}})^T \\ \bar{\bar{\beta}}^T & (\bar{\bar{\alpha}}^{-1})^T \end{bmatrix} = \bar{\bar{\Lambda}} \tag{5.2.33}$$

where the superscript T denotes the transpose of the matrix. This confirms that $\bar{\bar{\Lambda}}$ is a symmetric 6×6 matrix.

Additionally, we can demonstrate the following identities:

$$\bar{\bar{\Lambda}}^T \cdot \begin{bmatrix} \bar{\bar{I}} & \bar{\bar{0}} \\ \bar{\bar{0}} & -\bar{\bar{I}} \end{bmatrix} \cdot \bar{\bar{\Lambda}} = \begin{bmatrix} \bar{\bar{I}} & \bar{\bar{0}} \\ \bar{\bar{0}} & -\bar{\bar{I}} \end{bmatrix}, \tag{5.2.34}$$

$$\bar{\bar{\Lambda}}^T \cdot \begin{bmatrix} \bar{\bar{0}} & \bar{\bar{I}} \\ \bar{\bar{I}} & \bar{\bar{0}} \end{bmatrix} \cdot \bar{\bar{\Lambda}} = \begin{bmatrix} \bar{\bar{0}} & \bar{\bar{I}} \\ \bar{\bar{I}} & \bar{\bar{0}} \end{bmatrix}. \tag{5.2.35}$$

Lorentz Transformation of Maxwell Equations (cont.)

To verify these, recall that $\bar{\bar{\alpha}}^{-1}$ is symmetric and $\bar{\bar{\beta}}$ is skew-symmetric, $\bar{\bar{\beta}}^T = -\bar{\bar{\beta}}$, and that the two tensors commute, $\bar{\bar{\alpha}}^{-1} \cdot \bar{\bar{\beta}} = \bar{\bar{\beta}} \cdot \bar{\bar{\alpha}}^{-1}$, since both are built from $\vec{\beta}$ alone. Using $\bar{\bar{\Lambda}}^T = \bar{\bar{\Lambda}}$ from (5.2.33), the left-hand side of (5.2.34) is the block product

$$\gamma^2 \begin{bmatrix} \bar{\bar{\alpha}}^{-1} & \bar{\bar{\beta}} \\ -\bar{\bar{\beta}} & \bar{\bar{\alpha}}^{-1} \end{bmatrix} \begin{bmatrix} \bar{\bar{I}} & \bar{\bar{0}} \\ \bar{\bar{0}} & -\bar{\bar{I}} \end{bmatrix} \begin{bmatrix} \bar{\bar{\alpha}}^{-1} & \bar{\bar{\beta}} \\ -\bar{\bar{\beta}} & \bar{\bar{\alpha}}^{-1} \end{bmatrix}.$$

Carrying out the multiplication, the diagonal blocks are $\pm\gamma^2[(\bar{\bar{\alpha}}^{-1})^2 + \bar{\bar{\beta}}^2]$, while the off-diagonal blocks,

$$\gamma^2 \left[\bar{\bar{\alpha}}^{-1}\bar{\bar{\beta}} - \bar{\bar{\beta}}\,\bar{\bar{\alpha}}^{-1}\right],$$

vanish by commutativity. For the diagonal blocks, squaring (5.2.3) and using $(\vec{\beta}\vec{\beta})^2 = \beta^2\,\vec{\beta}\vec{\beta}$ gives

$$(\bar{\bar{\alpha}}^{-1})^2 = \bar{\bar{I}} + \left(\frac{1}{\gamma^2} - 1\right)\frac{\vec{\beta}\vec{\beta}}{\beta^2},$$

while (5.2.5) supplies $\bar{\bar{\beta}}^2 = \vec{\beta}\vec{\beta} - \beta^2\bar{\bar{I}}$. Adding these and multiplying by γ^2, the $\vec{\beta}\vec{\beta}$ terms cancel because

$$\frac{1-\gamma^2}{\beta^2} + \gamma^2 = \frac{1-\gamma^2+\gamma^2\beta^2}{\beta^2} = 0,$$

having used $\gamma^2\beta^2 = \gamma^2 - 1$, while the multiples of $\bar{\bar{I}}$ combine through $\gamma^2 - \gamma^2\beta^2 = 1$. Hence each diagonal block equals $\bar{\bar{I}}$, which establishes (5.2.34). The second identity (5.2.35) follows in the same way: the swap matrix $\begin{bmatrix} \bar{\bar{0}} & \bar{\bar{I}} \\ \bar{\bar{I}} & \bar{\bar{0}} \end{bmatrix}$ interchanges the roles of the blocks, so the new diagonal blocks are $\gamma^2[\bar{\bar{\beta}}\,\bar{\bar{\alpha}}^{-1} - \bar{\bar{\alpha}}^{-1}\bar{\bar{\beta}}] = \bar{\bar{0}}$ and the new off-diagonal blocks are $\gamma^2[(\bar{\bar{\alpha}}^{-1})^2 + \bar{\bar{\beta}}^2] = \bar{\bar{I}}$, reproducing the right-hand side of (5.2.35).
To apply these identities without unwieldy repeated block vectors, define

$$\mathcal{F} = \begin{bmatrix} \vec{E} \\ c\vec{B} \end{bmatrix}, \quad \mathcal{F}' = \bar{\bar{\Lambda}}\mathcal{F}, \quad G_- = \begin{bmatrix} \bar{\bar{I}} & \bar{\bar{0}} \\ \bar{\bar{0}} & -\bar{\bar{I}} \end{bmatrix}, \quad G_\times = \begin{bmatrix} \bar{\bar{0}} & \bar{\bar{I}} \\ \bar{\bar{I}} & \bar{\bar{0}} \end{bmatrix}.$$

Using (5.2.34) gives, step by step,

$$\mathcal{F}'^T G_- \mathcal{F}' = \mathcal{F}^T \bar{\bar{\Lambda}}^T G_- \bar{\bar{\Lambda}}\mathcal{F} = \mathcal{F}^T G_- \mathcal{F}. \tag{5.2.36}$$

Lorentz Transformation of Maxwell Equations (cont.)

Expanding the first and last quadratic forms in (5.2.36) yields

$$\vec{E}' \cdot \vec{E}' - c^2 \vec{B}' \cdot \vec{B}' = \vec{E} \cdot \vec{E} - c^2 \vec{B} \cdot \vec{B}. \tag{5.2.37}$$

Thus, at a fixed spacetime event, all inertial observers assign the same value to this scalar combination. It is invariant under a change of inertial frame, but it need not be constant from one spacetime event to another.
Likewise, (5.2.35) gives

$$\mathcal{F}'^T G_\times \mathcal{F}' = \mathcal{F}^T \bar{\bar{\Lambda}}^T G_\times \bar{\bar{\Lambda}} \mathcal{F} = \mathcal{F}^T G_\times \mathcal{F}. \tag{5.2.38}$$

The $G_\times$ quadratic form equals twice the corresponding dot product:

$$\mathcal{F}'^T G_\times \mathcal{F}' = 2\vec{E}' \cdot c\vec{B}', \qquad \mathcal{F}^T G_\times \mathcal{F} = 2\vec{E} \cdot c\vec{B}.$$

Cancelling the common factor of two yields the second Lorentz invariant:

$$\vec{E}' \cdot c\vec{B}' = \vec{E} \cdot c\vec{B}. \tag{5.2.39}$$

Further Reading

The discussion of Lorentz transformation of Maxwell equations largely follows Kong's dyadic formulation [17, 18], which aligns with the notation used in Chapter 3. Advanced tools such as tensor notation and four-vectors are omitted due to the scope of this course; fuller treatments of relativistic electrodynamics may be found in Zangwill [31] and in Jackson's classic text [13], which presents elegant derivations in Gaussian units. For a broader physical perspective on radiation—especially emission from accelerated charges and its connection to relativistic theory—Smith's pedagogical exposition [26] is recommended.

Problems

1. Consider the one-dimensional Lorentz transformation, show explicitly that $-(ct')^2 + (x')^2 + (y')^2 + (z')^2 = -(ct)^2 + x^2 + y^2 + z^2$.
2. Show (5.2.3) by direct multiplication.
3. Independently reproduce the derivation of the transformation laws for $\vec{D}$ and $\vec{H}$ by substituting (5.2.8) and (5.2.9) into Ampère–Maxwell's law ($\vec{\nabla} \times \vec{H} - \partial_{ct}(c\vec{D}) = \vec{J}$) and electric Gauss's law ($\vec{\nabla} \cdot c\vec{D} = c\rho$) in the S frame. Verify that the result agrees with (5.2.2).
4. Demonstrate that similar invariant quantities can be formed from $\vec{D}$ and $\vec{H}$ fields. Specifically, show that:

$$\vec{D}' \cdot \vec{H}' = \vec{D} \cdot \vec{H},$$
$$|c\vec{D}'|^2 - |\vec{H}'|^2 = |c\vec{D}|^2 - |\vec{H}|^2.$$

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